

AINSWORTH, NE, USA

WMO: 725556

Lat: 42.577N

Lon: 100.001W

Elev: 788

StdP: 92.21

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 94975

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.1	-17.9	-25.0	0.4	-20.6	-22.0	0.6	-17.5	15.4	-5.1	14.1	-4.3	4.1	320	0.650

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	35.1	20.6	32.8	20.6	31.1	20.3	23.5	29.7	22.6	28.8	21.7	27.9	5.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.1	18.4	25.8	21.0	17.2	25.4	19.9	16.1	24.9	74.4	29.7	70.7	28.8	67.2	28.0	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.4	11.7	10.5	DB	-25.7	37.8	2.5	2.6	-27.5	39.7	-29.0	41.2	-30.5	42.7	-32.3	44.6	(4)
(5)			WB	-25.7	24.9	2.3	0.9	-27.3	25.6	-28.6	26.1	-29.9	26.5	-31.6	27.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.4	-3.7	-2.5	2.9	8.2	13.9	19.6	23.7	22.1	17.6	9.7	3.0	-2.9	(6)	
	DBStd	11.12	7.14	7.07	6.86	5.60	4.86	3.90	3.45	3.28	4.83	5.48	6.27	6.51	(7)	
	HDD10.0	1820	424	351	234	97	18	1	0	0	3	71	221	400	(8)	
	HDD18.3	3703	683	583	479	306	152	31	3	7	71	270	460	658	(9)	
	CDD10.0	1591	0	1	14	42	140	290	425	376	230	63	11	0	(10)	
	CDD18.3	431	0	0	0	1	15	69	170	124	48	3	0	0	(11)	
	CDH23.3	4686	0	0	12	47	218	730	1847	1182	566	77	7	0	(12)	
	CDH26.7	1929	0	0	1	9	63	249	873	491	222	20	1	0	(13)	
(14) Wind	WSAvg	4.8	4.9	5.0	5.2	5.6	5.2	4.6	4.1	4.1	4.6	4.8	4.9	4.8	(14)	
(15) Precipitation	PrecAvg	558	10	15	28	69	92	100	74	54	48	40	17	12	(15)	
	PrecMax	716	33	55	67	162	181	207	139	97	136	92	53	38	(16)	
	PrecMin	295	1	2	3	16	12	17	14	5	10	1	0	1	(17)	
	PrecStd	106	7	11	15	32	47	43	37	22	29	27	14	9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.0	18.7	25.0	27.8	31.4	34.8	39.0	37.0	34.8	29.1	23.9	16.1	(19)	
		MCWB	6.5	8.3	11.4	13.7	18.3	20.9	20.4	20.6	18.2	15.4	11.0	7.4	(20)	
	2%	DB	11.3	13.7	21.0	24.0	28.1	31.7	36.2	33.7	32.1	25.2	18.9	11.7	(21)	
		MCWB	5.0	6.3	10.5	12.5	17.3	20.8	21.3	20.6	19.0	13.9	9.4	5.2	(22)	
	5%	DB	7.9	10.1	17.1	21.2	25.9	29.5	33.7	31.8	29.1	22.3	15.3	7.9	(23)	
		MCWB	3.2	4.4	9.1	11.6	16.3	19.8	21.3	20.6	18.3	13.0	7.4	3.2	(24)	
	10%	DB	5.5	7.1	13.0	17.8	23.0	27.6	31.8	29.1	27.0	18.9	12.2	5.2	(25)	
		MCWB	1.9	3.0	7.1	10.3	14.8	19.0	21.3	20.3	17.4	11.6	6.3	1.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.2	9.3	14.1	15.6	20.3	23.8	25.0	24.1	22.3	17.3	12.1	8.0	(27)	
		MCDB	14.5	16.8	18.5	23.6	27.6	29.5	31.1	29.9	28.4	23.9	20.1	14.7	(28)	
	2%	WB	5.2	6.7	11.4	13.8	18.8	22.3	23.9	22.9	20.7	15.7	10.0	5.4	(29)	
		MCDB	10.9	12.9	18.8	21.7	25.9	28.5	30.2	28.8	27.4	22.1	17.4	11.0	(30)	
	5%	WB	3.5	4.7	9.5	12.4	17.3	21.3	23.0	22.0	19.7	13.9	8.2	3.6	(31)	
		MCDB	7.6	9.8	16.9	19.5	23.8	27.3	29.5	27.7	26.3	20.2	15.3	7.9	(32)	
	10%	WB	2.0	3.0	7.4	10.9	15.9	20.1	22.1	21.2	18.6	12.2	6.3	1.8	(33)	
		MCDB	5.1	6.6	13.3	17.1	21.5	26.1	28.5	26.8	24.7	18.0	11.4	4.8	(34)	
(35) Mean Daily Temperature Range	MDBR		11.0	11.3	12.2	12.4	12.0	12.2	13.0	12.6	13.6	12.9	12.2	10.6	(35)	
	5% DB	MCDBR	14.5	15.8	18.6	18.3	16.2	15.0	16.2	16.1	17.1	18.2	17.4	14.4	(36)	
		MCWBR	9.4	9.3	9.7	9.1	7.5	6.5	5.5	5.9	6.5	8.7	9.2	9.3	(37)	
	5% WB	MCDBR	13.9	15.0	17.0	16.1	13.9	13.2	13.5	13.2	13.8	15.6	16.7	14.0	(38)	
		MCWBR	9.2	9.2	9.4	8.5	7.0	6.7	6.2	5.9	6.4	8.6	9.1	9.2	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.260	0.270	0.294	0.330	0.350	0.365	0.383	0.385	0.344	0.312	0.271	0.260	(40)	
	taud		2.528	2.510	2.484	2.386	2.417	2.406	2.393	2.366	2.445	2.507	2.544	2.513	(41)	
	Ebn at Noon		906	949	958	938	923	906	885	872	888	883	883	871	(42)	
	Edh at Noon		68	82	96	114	114	116	116	116	99	82	67	63	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.08	2.98	4.02	5.06	5.99	6.83	6.94	5.90	4.78	3.30	2.28	1.76	(44)	
	RadStd		0.16	0.20	0.34	0.29	0.48	0.44	0.24	0.34	0.30	0.34	0.17	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page